

QUARTIC KELLER MAPS AND LEADING CURVES AT INFINITY

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ABSTRACT. The known three-dimensional counterexample to the Jacobian conjecture has ordinary coordinate degree seven, while degree at most three is classically known to be invertible. This leaves ordinary degrees four, five, and six. For $F = LX + H_2 + H_3 + H_4$, the rescaling

$$K_\epsilon(X) = \epsilon^4 F(X/\epsilon)$$

satisfies $\det JK_\epsilon = \epsilon^9 \det JF$. Thus the Keller condition is a maximal-contact condition along the determinantal hypersurface, and the leading map collapses the projective boundary to a point or a rational curve.

Our first result excludes one-dimensional leading target span: if the three coordinate forms of F_4 span a line, then a degree-at-most-four Keller map $\mathbb{A}_{\mathbb{C}}^3 \rightarrow \mathbb{A}_{\mathbb{C}}^3$ is a polynomial automorphism. The proof reduces the transverse cubic pencil to a cube-containing or binary pencil and then uses two universal determinant syzygies.

This excludes in ordinary degree four the rank-one leading mechanism visible in the known degree-seven example. We also treat a conic-image stratum, in which

$$F_4 = G(x, y, z)(x^2, xy, y^2).$$

If the quadratic G is reduced and the base point $[0 : 0 : 1]$ is outside $G = 0$ or is a nonsingular point of it, then the Keller equations are inconsistent. The final step is an exact symbolic calculation in four parabolic orbits. We record separately a structural reduction for line image and the rational-curve strata that remain open.

Working manuscript. The exact symbolic calculations have been reproduced from the recovered sources. The pencil reductions and computer-assisted conic argument still require independent specialist review.

1. INTRODUCTION

For a polynomial map $F = (F_1, F_2, F_3)$, its *ordinary degree* is

$$\deg F = \max_i \deg F_i.$$

This convention matters: generic fiber degree and ordinary coordinate degree are different invariants.

The counterexample has ordered coordinate-degree profile $(7, 6, 4)$, and hence ordinary degree seven [Alp, Tao]. We credit Akhil Mathew as the primary human source of the problem: Mathew prompted Levent Alpöge, Alpöge prompted Fable, Fable produced the example, and Alpöge announced

Date: July 22, 2026.

it publicly [Ula]. Vistoli proved that every three-dimensional Keller map of ordinary degree at most three is an automorphism [Vis99]. Thus, if $D_{\min}$ denotes the least ordinary degree of a three-dimensional counterexample, the presently known endpoints give

$$4 \leq D_{\min} \leq 7.$$

The interval is background, not a novelty claim.

Write an affine-normalized degree-four map as

$$F = LX + H_2 + H_3 + H_4, \quad L \in \mathrm{GL}_3(\mathbb{C}), \quad (1.1)$$

with H_i homogeneous of degree i . Define the *leading target span*

$$\rho_4(F) = \dim_{\mathbb{C}} \langle (H_4)_1, (H_4)_2, (H_4)_3 \rangle \subset \mathrm{Sym}^4((\mathbb{C}^3)^\vee). \quad (1.2)$$

This is not the generic rank of JH_4 , Waring rank, or tensor rank.

There is a complementary geometric invariant. The projectivized leading map

$$\phi_4: \mathbb{P}^2 \dashrightarrow \mathbb{P}^2, \quad [x : y : z] \mapsto [H_{4,1} : H_{4,2} : H_{4,3}]$$

has image closure $C_4(F)$. The determinant equation implies that this image has dimension at most one. Subject to base-locus degeneracies, the cases $\rho_4 = 1, 2, 3$ correspond to point, line, and nondegenerate rational-curve image. We use ρ_4 for exact linear algebra and $C_4(F)$ to organize the geometry at infinity.

Our main theorem excludes $\rho_4(F) = 1$.

Theorem 1.1 (One-dimensional leading target span). *Let $F: \mathbb{A}_{\mathbb{C}}^3 \rightarrow \mathbb{A}_{\mathbb{C}}^3$ be a Keller map of ordinary degree at most four. If $\rho_4(F) = 1$, then F is a polynomial automorphism.*

The theorem allows arbitrary mixed quadratic and cubic layers. It assumes neither that $F - X$ is homogeneous nor that its Jacobian is nilpotent. This distinguishes it from the homogeneous three-dimensional theorem of de Bondt–van den Essen [dBvdE05]. General invariant theory of binary quartic pencils is classical [Wal98]; our claim concerns the restrictions imposed by the mixed homogeneous Keller equations.

Recent work restricts graded Keller maps [Sha26]. We impose no torus action. As of July 23, 2026, we know no public theorem excluding every ordinary-degree-four three-dimensional Keller map.

Our second theorem concerns a conic-image leading form rather than linear target span two.

Theorem 1.2 (Four reduced conic orbits). *Suppose a degree-four Keller map has, after independent linear source and target changes,*

$$H_4 = G(x, y, z)(x^2, xy, y^2),$$

where G is a reduced quadratic form. Put $p = [0 : 0 : 1]$. If $p \notin V(G)$, or if p is a nonsingular point of $V(G)$, then no such Keller map exists.

The final coefficient elimination in theorem 1.2 is computer-assisted. The calculation keeps an arbitrary invertible linear part and never divides by a parameter. The exact scope is described in section 9.

2. KELLER JETS AT INFINITY

The homogeneous determinant identities have a compact geometric source. Define

$$K_\epsilon(X) = \epsilon^4 F(X/\epsilon) = H_4 + \epsilon H_3 + \epsilon^2 H_2 + \epsilon^3 L X.$$

Since $JK_\epsilon = \epsilon^3 JF(X/\epsilon)$,

$$\det JK_\epsilon = \epsilon^9 \det JF. \quad (2.1)$$

In particular, $\det JH_4 = 0$. More strongly, every coefficient below the maximum possible order nine in the determinant arc

$$JH_4 + \epsilon JH_3 + \epsilon^2 JH_2 + \epsilon^3 L$$

vanishes. Thus a quartic Keller map is a maximally volume-preserving deformation of a singular leading map at infinity.

The rank-one theorem treats the case in which $C_4(F)$ is a point. The span-two reduction treats line image. The conic theorem begins the first nonlinear case. This explains why the cases belong in one paper: they are successive image geometries of the same boundary jet, rather than unrelated coefficient ansatzes.

Remark 2.1 (Determinantal arcs). Equation (2.1) says that the matrix arc JK_ϵ has maximal contact with the determinant hypersurface in $\text{Mat}_{3 \times 3}$. Our matrices are not arbitrary arcs: each coefficient is the Jacobian of a homogeneous polynomial map. The integrability constraint is what turns a coarse determinantal problem into the pencil, coordinate, and first-integral geometry used below.

3. TWO COORDINATE LEMMAS

We work temporarily over an algebraically closed field k of characteristic zero.

Lemma 3.1 (Quadratic coordinate). *If $f \in k[X_1, X_2, X_3]$ has degree at most two and no critical point, then some constant directional derivative of f is nonzero. In particular, f is a coordinate.*

Remark 3.2 (Sharpness of low-degree coordinate rigidity). The quadratic hypothesis is essential. The polynomial

$$f = x + x^2 y$$

has no critical point, but its zero fiber is reducible, so it is not a coordinate. Likewise $x^4 + x + x^2 y$ has pure-power quartic top part, no critical point, and a reducible zero fiber. The coordinate lemmas therefore encode a genuine low-degree threshold rather than an unrestricted submersion principle.

Proof. Write

$$f = \frac{1}{2}X^TAX + b^TX + c$$

with A symmetric. If $b \in \text{im } A$, the equation $AX + b = 0$ has a solution, contrary to the hypothesis. Hence

$$b \notin \text{im } A = (\ker A)^\perp.$$

There is $u \in \ker A$ with $b^Tu \neq 0$, and $D_u f = b^Tu$. After making u a coordinate direction, a triangular change replaces that coordinate by f . $\square$

Lemma 3.3 (Cubic-cube coordinate). *Suppose that $\deg f \leq 3$, f has no critical point, and its cubic homogeneous part is $\lambda \ell^3$ with $\lambda \neq 0$. Then f has a nonzero constant directional derivative and is a coordinate. The straightening and its inverse can be chosen of degree at most three.*

Proof. Take $\ell = x$, write the remaining variables as $w \in k^2$, and express

$$\nabla_w f = Mw + ax + b$$

with M symmetric. Project this affine expression to $\text{coker } M = (\ker M)^\vee$.

If it never vanishes, the two projected vectors a, b do not span an affine line through the origin. Hence some $u \in \ker M$ satisfies $u^Ta = 0$ and $u^Tb \neq 0$, so $D_u f$ is a nonzero constant.

Otherwise fix a solution of $Mw + ax + b = 0$. If a pairs nontrivially with $\ker M$, variation within the solution space makes f_x vanish. If it does not, then $a \in \text{im } M$; solvability for one x also puts b in $\text{im } M$, so the system is solvable for every x . Substitution of a linear solution $w = w(x)$ makes f_x a quadratic polynomial in x with leading coefficient 3λ . It has a root over k , again producing a critical point. Both alternatives contradict the hypothesis, so the constant-direction case is forced.

In suitable linear coordinates $f = cw_2 + \psi(x, w_1)$, with $c \neq 0$ and $\deg \psi \leq 3$. Replacing w_2 by f gives the asserted straightening and degree bound. $\square$

Proposition 3.4 (The easy transverse branches). *In (1.1), suppose*

$$H_4 = (0, 0, h), \quad H_3 = (P, Q, R).$$

If P, Q are linearly dependent, or if their pencil contains the cube of a linear form, then F is an automorphism.

Proof. A row of JF cannot vanish, so every target combination of the coordinates of F has no critical point. In the dependent case a target combination of the first two components has degree at most two, and lemma 3.1 applies. In the cube-containing case a combination has degree at most three with cubic part a cube, and lemma 3.3 applies.

After straightening this component to a variable t , choose the other target combination among the first two original components. Over $\overline{k(t)}$, the remaining two coordinates form a plane Keller map. In the quadratic case one component has degree at most six. In the cubic case it has degree in

$$\{1, 2, 3, 4, 5, 6, 7, 9\}.$$

Indeed, substituting $w_2 = c^{-1}(t - \psi)$ into a cubic gives degree at most seven unless the w_2^3 term and the cubic part of ψ are both nonzero, in which case the unique top term has degree nine. Every displayed integer is a product of at most two primes, counted with multiplicity. The corrected Appelgate–Onishi plane theorem, in the form completed by Nagata, therefore makes the plane map an automorphism [AO85, Nag88]. It follows that F is birational. A birational Keller self-map is injective by Zariski’s Main Theorem and hence an automorphism by Ax–Grothendieck. $\square$

4. THE RATIONAL-COMPOSITE PENCIL LEMMA

The highest nonautomatic homogeneous Keller equation for the normalization in proposition 3.4 is

$$dP \wedge dQ \wedge dh = 0. \quad (4.1)$$

If P, Q are independent, they are algebraically independent and (4.1) makes h algebraic over $k(P, Q)$.

Lemma 4.1 (Weighted one-variable field). *Let A, B be coprime homogeneous forms of the same degree n , put*

$$t = A/B, \quad w = h^3/B^4,$$

and suppose that h is algebraic over $k(A, B)$. Then w is algebraic over $k(t)$.

If $K_1 = k(t, w)$, there are coprime homogeneous forms A_0, B_0 of a common degree d and a rational function $\mathcal{R}$ of degree e such that

$$K_1 = k(A_0/B_0), \quad t = \mathcal{R}(A_0/B_0), \quad n = ed.$$

Proof. Choose a polynomial relation among A, B, h and retain a nonzero source-homogeneous part. With weights 3, 3, 4, all exponents of h in that part are congruent modulo three. After removing a common power of h , the relation has the form

$$\sum_i C_i(A, B) h^{3i} = 0, \quad \deg C_i = M - 4i.$$

Division by B^M gives a polynomial relation for w over $k(t)$.

The one-variable field $K_1 \subset k(\mathbb{P}^2)$ is the function field of a curve dominated by $\mathbb{P}^2$. Restriction to a suitable line makes that curve unirational, hence rational in characteristic zero. Thus $K_1 = k(s)$, and s can be written A_0/B_0 with A_0, B_0 coprime homogeneous forms of the same degree d .

Write $\mathcal{R} = U/V$, where U, V are coprime binary forms of degree e . The substituted forms $U(A_0, B_0)$ and $V(A_0, B_0)$ remain coprime: their common divisor would, by the binary resultant identities, divide powers of both A_0 and B_0 . Comparing degrees in the reduced ratio $A/B = \mathcal{R}(A_0/B_0)$ yields $n = ed$. $\square$

Proposition 4.2 (Cubic pencils). *Let P, Q be independent homogeneous cubics satisfying (4.1). Then their pencil either contains the cube of a linear form, or P, Q, h are polynomials in two independent linear forms.*

Proof. First suppose $\gcd(P, Q) = 1$. Apply lemma 4.1 with $A = P, B = Q$. If $e > 1$, then $3 = ed$ gives $e = 3, d = 1$, so P, Q are binary in two linear forms.

If $e = 1$, write $w = R(t)$. For $F_\xi = P - \xi Q$, let $r_\xi = \text{ord}_\xi R$, and set

$$c_\xi = r_\xi + 4\mathbf{1}_{\xi=\infty}.$$

If an irreducible component of F_ξ has multiplicity m , valuation gives

$$3\nu(h) = c_\xi m.$$

Every c_ξ is nonnegative and

$$\sum_{\xi \in \mathbb{P}^1} c_\xi = 4.$$

If no fiber is a cube, every cubic fiber has a component whose multiplicity is prime to three. The displayed valuation then makes every c_ξ a multiple of three, contradicting their sum.

Now write $P = GA, Q = GB$ with $G \neq 1$ and $\gcd(A, B) = 1$. If $\deg G = 2$, then A, B are independent linear forms. An irreducible factor of G , of multiplicity $m = 1$ or 2 , cannot carry a nonconstant A/B , since its valuation would give

$$3\nu(h) = 4m.$$

Every component of G is therefore a fiber line of the pencil $\langle A, B \rangle$; hence G, P, Q are binary.

If $G = \ell$, the reduced pencil A/B has degree two. In the composite case lemma 4.1 gives $e = 2, d = 1$. The valuation along ℓ forces ℓ into the same binary pencil. In the primitive case, the same valuation first makes A/B constant on $\ell = 0$. After changing the pencil basis, write $A = \ell m$. If m is proportional to ℓ , then $P = \ell^3$. Otherwise, with $r = \text{ord}_0 R$, valuations along $m = 0$ and $\ell = 0$ give

$$3\nu_m(h) = r, \quad 3\nu_\ell(h) = r + 4,$$

which is impossible modulo three.

Finally, if P, Q are binary in linear forms a, b , then

$$dP \wedge dQ = J_{a,b}(P, Q) da \wedge db$$

with nonzero planar Jacobian. Equation (4.1) forces the derivative of h transverse to a, b to vanish, so h is binary in the same forms. $\square$

5. PROOF OF THE ONE-DIMENSIONAL-SPAN THEOREM

By a target change, write $H_4 = (0, 0, h)$. In the branch not already covered by proposition 3.4, proposition 4.2 gives coordinates such that

$$P, Q \in k[x, y]_3, \quad h \in k[x, y]_4,$$

and the pencil $\langle P, Q \rangle$ contains no cube.

Put

$$W = J(P, Q), \quad U = J(P, h), \quad V = J(Q, h),$$

where J is the planar Jacobian in x, y . Write every possible z -dependent lower term as

$$\begin{aligned}(H_2)_1 &= A_0(x, y) + za(x, y) + \alpha z^2, \\ (H_2)_2 &= B_0(x, y) + zb(x, y) + \beta z^2, \\ (H_2)_3 &= C_0(x, y) + zr(x, y) + cz^2, \\ (H_3)_3 &= R_0(x, y) + zq(x, y) + z^2\ell(x, y) + \rho z^3,\end{aligned}\tag{5.1}$$

where a, b, r, ℓ are linear and q is quadratic.

Introduce T and expand

$$\det(L + TJH_2 + T^2JH_3 + T^3JH_4) = \det L + \sum_{j=1}^7 T^j E_j.$$

Exact expansion gives

$$[z^2]E_6 = 3\rho W, \tag{5.2}$$

$$[z]E_6 = 2(W\ell + \alpha V - \beta U), \tag{5.3}$$

$$[1]E_6 = Wq + Va - Ub, \tag{5.4}$$

$$[z^3]E_5 = -2J(\ell, \alpha Q - \beta P), \tag{5.5}$$

where the last three identities are read after $\rho = 0$. After $\alpha = \beta = \ell = \rho = 0$, one also has

$$\begin{aligned}[z]E_5 &= -4J(a, b)h - qJ(a, Q) + 3QJ(a, q) \\ &\quad + qJ(b, P) - 3PJ(b, q) + 2cW,\end{aligned}\tag{5.6}$$

$$[z^2]E_4 = -J(a, b)q + 2c(J(a, Q) - J(b, P)), \tag{5.7}$$

$$[z^3]E_3 = 2cJ(a, b). \tag{5.8}$$

Lemma 5.1 (First binary syzygy). *In the cube-free branch,*

$$\rho = \ell = \alpha = \beta = 0.$$

Proof. Equation (5.2) gives $\rho = 0$. Put $S = \alpha Q - \beta P$. Equations (5.3) and (5.5) say

$$W\ell + J(S, h) = 0, \quad J(\ell, S) = 0.$$

If $\ell \neq 0$, the second equation makes $S = \lambda\ell^3$. Cube-freeness gives $S = 0$, whence $\alpha = \beta = 0$, and the first equation becomes $W\ell = 0$, a contradiction.

Thus $\ell = 0$, and $J(S, h) = 0$. Algebraically dependent homogeneous binary forms of coprime degrees three and four are powers of a common linear form. A nonzero S would therefore be a cube in $\langle P, Q \rangle$. Hence $S = 0$, and independence of P, Q gives $\alpha = \beta = 0$. $\square$

Lemma 5.2 (Second binary syzygy). *In the cube-free branch,*

$$a = b = q = c = 0.$$

Proof. Let $D = J(a, b)$. If $D \neq 0$, equations (5.8), (5.7), and (5.6), in that order, give $c = 0, q = 0, h = 0$, a contradiction. Thus $D = 0$.

If $a = b = 0$, equations (5.4) and (5.6) give $q = c = 0$. Otherwise a row operation on the first two target coordinates makes $b = 0, a \neq 0$. If $c \neq 0$, (5.7) makes $J(a, Q) = 0$, so Q is a cube. Hence $c = 0$.

Equation (5.6) is now

$$-qJ(a, Q) + 3QJ(a, q) = 0.$$

If $q = 0$, equation (5.4) gives $J(Q, h) = 0$, again making Q a cube. Otherwise take $a = x$; then

$$qQ_y - 3Qq_y = 0, \quad \partial_y(Q/q^3) = 0.$$

Homogeneity gives $x^3Q = \lambda q^3$. Unique factorization makes Q a cube, the final contradiction. $\square$

Proof of theorem 1.1. The two syzygy lemmas reduce the map to

$$F_1 = f_1(x, y) + \gamma z, \quad F_2 = f_2(x, y) + \phi z, \quad F_3 = f_3(x, y) + (\mu + r(x, y))z,$$

with r linear. The coefficient of z in the determinant is

$$J(r, \phi f_1 - \gamma f_2).$$

If $(\gamma, \phi) \neq (0, 0)$ and $r \neq 0$, its vanishing makes the nonzero cubic $\phi P - \gamma Q$ a polynomial in the single linear form r , hence a cube. Therefore $r = 0$. A target linear change sends the nonzero vector (γ, ϕ, μ) to e_3 , and the map becomes

$$(G_1(x, y), G_2(x, y), z + G_3(x, y)).$$

The plane pair is Keller and one coordinate has degree at most four, so [AO85] makes it an automorphism.

If $\gamma = \phi = 0$, then

$$\det JF = J(f_1, f_2)(\mu + r).$$

Both factors are units, so $r = 0$, and the same triangular conclusion holds. $\square$

6. A LEADING-TARGET-SPAN-TWO REDUCTION

For context, suppose instead that

$$H_4 = (P, Q, 0)$$

with P, Q algebraically independent quartics, and let R be the third cubic component of H_3 . The highest Keller equation is

$$\text{Jac}(P, Q, R) = 0.$$

Write $P = GA, Q = GB$, with $\gcd(A, B) = 1$, and set

$$n = \deg A = \deg B = 4 - \deg G, \quad t = A/B, \quad w = R^4/Q^3.$$

The proof of lemma 4.1, with weights 4, 4, 3, gives a rational intermediate field and a factorization $n = ed$.

Proposition 6.1 (Four structural loci). *Every such leading pair belongs to at least one of the following loci:*

- (i) *a binary quartic pencil, with R binary in the same two linear forms;*
- (ii) *a quadratic-source pencil*

$$P = U(a_2, b_2), \quad Q = V(a_2, b_2),$$

where a_2, b_2 are coprime quadrics and U, V are binary quadratics;

- (iii) *a primitive coprime pencil containing a fourth power ℓ^4 ;*
- (iv) *a primitive pencil whose fixed components are supported on special fibers.*

The loci need not be disjoint.

Proof. If the intermediate extension has degree $e > 1$, the possibilities $n = ed$ are

$\deg G$	n	(e, d)
0	4	$(4, 1), (2, 2)$
1	3	$(3, 1)$
2	2	$(2, 1)$.

The $d = 1$ cases are binary after valuation along every component of G ; the only nonbinary composite case is $(e, d) = (2, 2)$.

In the primitive coprime case $w = \rho(t)$. If a component of multiplicity m lies in the fiber $P - \xi Q$, then

$$4\nu(R) = c_\xi m, \quad c_\xi \geq 0, \quad \sum_{\xi \in \mathbb{P}^1} c_\xi = 3,$$

with the denominator contribution incorporated at infinity. Some c_ξ is odd, so every component multiplicity in that fiber is divisible by four. The quartic fiber is therefore ℓ^4 . With fixed components, the same valuation argument places every component not already binary over a special fiber, giving (iv). $\square$

Remark 6.2. Proposition 6.1 is a structural reduction, not an exclusion theorem. Its relation to the classical stratification of binary quartic pencils [Wal98] is that the Keller equation supplies the cubic R and the valuation restrictions; it does not make the ambient pencil theory new.

7. THE FOUR REDUCED CONIC ORBITS

We now turn to the separate conic-image form

$$H_4 = G(x, y, z)(x^2, xy, y^2). \tag{7.1}$$

Put $p = [0 : 0 : 1]$.

Lemma 7.1 (Parabolic orbit list). *Up to a source transformation fixing p and rescaling G , the seven orbits of pairs consisting of a plane conic and p*

have representatives

<i>geometry</i>	<i>G</i>
<i>smooth, $p \notin V(G)$</i>	$z^2 + xy$
<i>smooth, $p \in V(G)$</i>	$xz + y^2$
<i>two lines, $p \notin V(G)$</i>	$z^2 + x^2$
<i>two lines, p nonsingular on the union</i>	xz
<i>two lines, p the node</i>	xy
<i>double line, $p \notin V(G)$</i>	z^2
<i>double line, $p \in V(G)$</i>	x^2 .

Proof. The subgroup preserving the pencil $\langle x, y \rangle$ is the projective stabilizer of p . Over $\mathbb{C}$, plane conics are classified by rank. For rank three, the stabilizer is transitive on points on the conic and on points off it. For rank two, a point is off the union, on exactly one component, or at the node. For rank one, it is on or off the doubled line. Elementary linear changes give the displayed representatives. $\square$

For (7.1), define

$$\delta_G = xG_z\partial_x + yG_z\partial_y + (zG_z - 4G)\partial_z.$$

This is a right-kernel derivation for JH_4 . Let

$$n = (y^2, -2xy, x^2)^T.$$

Then n is a left-kernel vector, and

$$\delta_G(n) = 2G_z n.$$

Lemma 7.2 (Invariant fields). *For each of*

$$G = z^2 + xy, \quad xz + y^2, \quad z^2 + x^2, \quad xz,$$

one has

$$\ker_{\mathbb{C}(x,y,z)} \delta_G = \mathbb{C} \left(\frac{x}{y}, Gy^2 \right).$$

Consequently a homogeneous invariant rational function has scalar degree divisible by four.

Proof. Put $r = x/y$ and $s = Gy^2$. Direct differentiation gives $\delta_G(r) = \delta_G(s) = 0$.

For $G = xz + \epsilon y^2$, the full field is $\mathbb{C}(r, s)(y)$, since

$$z = \frac{s - \epsilon y^4}{ry^3}.$$

The induced nonzero derivation on this one-variable rational field has constant field $\mathbb{C}(r, s)$.

For $G = z^2 + xy$ and $z^2 + x^2$, put $w = yz$. Over $\mathbb{C}(r, s)$, the full field is the function field of

$$w^2 + ry^4 = s \quad \text{or} \quad w^2 + r^2y^4 = s.$$

Each generic curve is geometrically integral. Thus $\mathbb{C}(r, s)$ is relatively algebraically closed in the one-variable function field, and a nonzero $\mathbb{C}(r, s)$ -derivation has no further constants.

Under scalar dilation, r has degree zero and s degree four. The rational eigenvectors in $\mathbb{C}(r, s)$ therefore have degrees in $4\mathbb{Z}$. $\square$

Write $H_3 = (P, Q, R)$ and $M = n \cdot H_3$. The first normal Keller equation is

$$n^T JH_3 \delta_G = 0,$$

or

$$\delta_G(M) = 2G_z M.$$

Hence M/y^2 is invariant of degree three. By lemma 7.2, $M = 0$, and the syzygies of $(y^2, -2xy, x^2)$ give

$$H_3 = T(A, B) := (2xA, yA + xB, 2yB) \quad (7.2)$$

for quadratic forms A, B .

Introduce

$$K_\epsilon = H_4 + \epsilon H_3 + \epsilon^2 H_2 + \epsilon^3 L X, \quad \Phi(Y) = Y_1 Y_3 - Y_2^2.$$

The chain rule gives

$$\text{Jac}(K_{\epsilon,1}, K_{\epsilon,2}, \Phi(K_\epsilon)) = K_{\epsilon,1} \det JK_\epsilon. \quad (7.3)$$

For a Keller map,

$$\det JK_\epsilon = \epsilon^9 \det L.$$

Write

$$\Phi(K_\epsilon) = \epsilon^2 S_2 + \epsilon^3 S_3 + \epsilon^4 S_4 + \epsilon^5 S_5 + \epsilon^6 S_6.$$

The first two coefficients of (7.3) make S_2, S_3 δ_G -invariant. Their degrees are six and five, so lemma 7.2 gives

$$S_2 = S_3 = 0.$$

Now

$$S_2 = G n \cdot H_2 - (yA - xB)^2.$$

Since G is squarefree,

$$yA - xB = G\ell, \quad n \cdot H_2 = G\ell^2 \quad (7.4)$$

for a linear form $\ell = ax + by$. Every solution of the second equation is

$$H_2 = G(b^2, -ab, a^2) + T(U, V),$$

where U, V are linear. The first equation has the exhaustive solutions

G	A	B
$z^2 + xy$	$xG + ax^2 + bz^2$	$yG - az^2 - by^2$
$xz + y^2$	$xG + axy + bG$	$yG - axz$
$z^2 + x^2$	$xG + bG$	$yG - aG$
xz	$xG + bG$	$yG - aG,$

with C linear.

Proposition 7.3 (Exact terminal elimination). *For each of the four displayed G , the equations*

$$S_3 = 0, \quad \delta_G(S_4) = 0$$

and one further coefficient of (7.3) force $\det L = 0$.

Computer-assisted proof. Substitute the preceding parametrizations and retain all nine entries of L . Solving the linear equation $S_3 = 0$ leaves two entries of L free. The coefficients of $\delta_G(S_4) = 0$ are then solved without inverting a parameter. The orbit $G = xz + y^2$ has the exhaustive split

$$a = 0 \quad \text{or} \quad [z]C = [x]C.$$

In the resulting five charts, the next nonzero coefficient and the determinant have the form

G	required zero coefficient	$\det L$
$z^2 + xy$	$x^2 z G(A_0 x + 2B_0 y)^2$	$c_2(-bA_0 + 2aB_0)^2$
$xz + y^2, c_2 = c_0$	$\frac{1}{2}x^3 G(A_0 x - 2my)^2$	$c_2(bA_0 + 2am)^2$
$xz + y^2, a = 0$	$\frac{1}{2}x^3 G(nx + (4b^2(c_2 - c_0) - 2m)y)^2$	$b^2 c_2 n^2$
$z^2 + x^2$	$x^2 z G(A_0 x + B_0 y)^2$	$c_2(bA_0 - aB_0)^2$
xz	$\frac{1}{2}x^3 G(A_0 x + B_0 y)^2$	$c_2(bA_0 - aB_0)^2$.

Here the displayed letters are the residual scalar parameters after the linear solves. In every row, vanishing of the middle polynomial forces the squared linear form to vanish coefficientwise, and the last column then vanishes.

The accompanying assertion-based checker constructs the parametrizations, performs the linear solves, verifies the exhaustive branch equation, expands (7.3), and asserts each factorization exactly over $\mathbb{Q}$. The evidence boundary is stated in section 9. $\square$

Proof of theorem 1.2. Lemma 7.1 reduces the hypothesis to the four forms treated in proposition 7.3. That proposition contradicts the required invertibility of L . $\square$

Remark 7.4 (The three untreated conic orbits). The remaining representatives are

$$G = xy, \quad G = z^2, \quad G = x^2.$$

For $G = xy$ and $G = x^2$, the characteristic derivation has odd-degree polynomial invariants, so the first normal deformation need not vanish. For $G = z^2$, degree-six invariants survive, so S_2 need not vanish. The proof above does not specialize to these cases.

8. WHY THE DEGREE-SEVEN EXAMPLE ESCAPES

The public degree-seven map has a rank-one top layer, so it is important to say exactly why theorem 1.1 does not continue formally to degree seven. In the displayed coordinates, one transverse component is

$$R = 2x - 3x^2y - x^3z = x(2 - 3xy - x^2z).$$

It is a polynomial submersion:

$$R_y = -3x^2, \quad R_z = -x^3,$$

so a critical point would have $x = 0$, where $R_x = 2$. Nevertheless R is not a coordinate, because the fiber $R^{-1}(0)$ is reducible.

In ordinary degree four, the dependent transverse branch in the rank-one argument produces a quadratic polynomial submersion, and lemma 3.1 forces it to be a coordinate. The known counterexample reaches a quartic submersion that is smooth but globally nontrivial. Thus the degree gap is not an artifact of the proof: low-degree differential regularity still forces algebraic triviality, whereas degree seven has enough room for nontrivial behavior at infinity.

9. EXACT COMPUTATIONS AND REMAINING PROOF OBLIGATIONS

The recovered supplement contains one universal rank-one script and nine conic-orbit scripts. All ten programs were rerun with Python 3.12.11 and SymPy 1.14.0. Every new log is byte-for-byte identical to the stored output. The reviewed copies, stored outputs, and integrity manifests are in `code/rank-one/` and `code/recovered-conic/`.

For the conic theorem, the standalone checker

`code/verify_conic_certificates.py`

reconstructs the five terminal charts from the normal equations. It asserts the cubic and quadratic normal forms, $S_2 = S_3 = 0$, $\delta_G(S_4) = 0$, the set-theoretically exhaustive split $a = 0$ or $[z]C = [x]C$, and all five pairs of factorizations in proposition 7.3. It uses exact rational arithmetic, makes no random specializations, and divides by no parameter.

The rank-one script keeps arbitrary binary cubics P, Q , an arbitrary binary quartic h , every lower term in (5.1), and an arbitrary 3×3 linear part. It asserts (5.2)–(5.8) and the terminal determinant identity. It does not prove lemmas 3.3 and 4.1 and proposition 4.2.

The bounded equivariant calculation has a separate source directory. Its core scripts recompute the degree-four, degree-five, and degree-six eliminations over $\mathbb{Q}$, together with the exact target-orbit minor. Exploratory finite-field searches are kept in a visibly separate subdirectory and are not used as characteristic-zero nonexistence proofs. The computation index records which script supports each computer-assisted step.

The checker makes the computational implication in proposition 7.3 explicit. It does not certify lemmas 7.1 and 7.2; those are conventional arguments in the manuscript. A submission release therefore requires:

- (1) an independent audit of the rational-composite-pencil argument;
- (2) a second check of the parabolic orbit list and invariant fields; and
- (3) specialist review of the computer-assisted implication in proposition 7.3.

10. OPEN DEGREE-FOUR STRATA

Theorem 1.1 shows that an ordinary-degree-four counterexample must have leading target span at least two. Theorem 1.2 eliminates four reduced conic-image orbits. Neither result settles ordinary degree four. In particular, the following remain:

- the three degenerate conic orbits $G = xy, z^2, x^2$;
- the quadratic-source-pencil locus in proposition 6.1;
- primitive pencils with a fourth-power fiber or fixed components; and
- leading forms of full target span outside the conic-image model.

The natural next question is whether the surviving strata can be reduced to finitely many invariant-theoretic charts with exact coefficient certificates, or whether one of them supports a genuinely new quartic mechanism.

APPENDIX A. FIXED-FACTOR CONIC STRATA IN DEGREES FIVE AND SIX

This appendix pushes the conic leading-image method beyond degree four. It does not classify degrees five or six: the theorem isolates an open basepoint boundary and a resonant primitive sextic case.

Let the leading homogeneous map have the squarefree fixed-factor form

$$H_D = G(A^2, AB, B^2),$$

$$\deg G = g > 0, \quad \deg A = \deg B = e, \quad D = g + 2e.$$

Assume that A, B are a regular sequence and that some irreducible component of $G = 0$ sees a nonconstant ratio A/B . The latter is the basepoint-free invariant-gap locus used below.

Lemma A.1 (Invariant-degree gap). *For*

$$\Omega = A dG \wedge dB - B dG \wedge dA + 2G dA \wedge dB,$$

let δ be the associated homogeneous derivation. Every nonzero homogeneous P satisfying $\delta(P) = 0$ has degree divisible by D .

Proof. Choose an irreducible factor $\Gamma \mid G$, of degree h , on which A/B is nonconstant, and write $P = \Gamma^r Q$ with $\Gamma \nmid Q$. On the function field of $\Gamma = 0$, both the induced derivation and the Euler derivation annihilate A/B ; hence $\bar{\delta} = cE$ for a nonzero rational function c . From $\delta(GB^2) = 0$ one gets

$$\overline{\delta\Gamma/\Gamma} = -c(D - h).$$

Reducing $\delta(\Gamma^r Q) = 0$ modulo Γ now yields

$$c(\deg P - rD)\bar{Q} = 0.$$

Thus $\deg P = rD$. □

Write a Keller jet as

$$K_\varepsilon = H_D + \varepsilon H_{D-1} + \cdots + \varepsilon^{D-2} H_2 + \varepsilon^{D-1} LX$$

and set

$$\Phi(Y) = Y_1 Y_3 - Y_2^2, \quad \Phi(K_\varepsilon) = \sum_{j \geq 1} \varepsilon^j S_j.$$

The chain-rule identity

$$\text{Jac}(K_{\varepsilon,1}, K_{\varepsilon,2}, \Phi(K_\varepsilon)) = K_{\varepsilon,1} \det JK_\varepsilon$$

and lemma A.1 imply inductively that

$$S_1 = \cdots = S_{D-1} = 0.$$

Indeed, $\deg S_j = 2D - j$, which lies strictly between D and $2D$.

Theorem A.2 (Basepoint-free fixed-factor conic exclusion). *Under the preceding hypotheses, no nonautomorphic Keller map has one of the following leading-form types:*

$$\begin{array}{l|l} D = 5 & (g, e) = (1, 2), (3, 1) \\ D = 6 & (g, e) = (2, 2), (4, 1). \end{array}$$

Proof idea. The vanishings $S_1, \dots, S_{D-1}$ combine with the Koszul syzygies of (A, B) and the Hilbert–Burch syzygies of $(A, B)^2$. They lift the deformed rank-one symmetric matrix to

$$\widehat{M}_\varepsilon = a_\varepsilon P_\varepsilon P_\varepsilon^T + \varepsilon^3 \text{sym}(P_\varepsilon, Q_\varepsilon)$$

after a source-independent target correction. Consequently

$$\Phi(\widehat{K}_\varepsilon) = -\varepsilon^6 (P_\varepsilon \wedge Q_\varepsilon)^2.$$

The chain-rule identity forces every non- ε irreducible factor of $P_\varepsilon \wedge Q_\varepsilon$ to divide both of the first two coordinates and then the Keller determinant. This is impossible. The remaining possibility is a pure power of ε ; comparison of the lowest coefficients makes the wedge vanish and again forces the linear determinant to be zero.

Under `code/degree-five-six/`, the file `conic_fixed_factor_d5_d6_proof_note.md` writes out the four graded lift identities. The script `verify_conic_hensel_d5_d6.py` replays them exactly. $\square$

The $(g, e) = (3, 1)$ quintic case admits a stronger boundary analysis.

Theorem A.3 (Three-variable cubic-factor quintics). *Let G be a squarefree cubic genuinely depending on x, y, z . There is no Keller jet with leading form*

$$G(x, y, z)(x^2, xy, y^2).$$

Thus the only unresolved cubic-factor overlap of this shape is the binary case $G \in \mathbb{C}[x, y]$.

Proof structure. At the base point $[0 : 0 : 1]$, the four conic normal equations split according to the multiplicity of $G = 0$ and whether the first normal linear form vanishes there. The unit branch is eliminated by the rank of the tangent cone and, in the rank-one case, by two incompatible coefficients. In the regular branch a source translation removes the first syzygy. The

remaining cusp, node, transverse line–conic, and tangent line–conic normal forms each collapse under two or three displayed coefficients of the Keller determinant. Multiplicity three forces G to be binary.

The complete normal-form ledger and decisive coefficients are recorded in `quintic_cubic_factor_basepoints.md`, under `code/degree-five-six/`. The companion exact script replays every coefficient identity used in the case table. $\square$

Question A.4 (Binary cubic-factor overlap). *Can a quintic Keller map have leading form*

$$G(x, y)(x^2, xy, y^2)$$

for a squarefree binary cubic G ?

Question A.5 (Fixed-factor basepoints in degree five). *Can the remaining $(g, e) = (1, 2)$ quintic branches, where the base locus of (A, B) meets $G = 0$, satisfy the Keller equations?*

Question A.6 (Primitive sextic conic stratum). *Can a degree-six Keller map have primitive conic leading form*

$$(A_3^2, A_3B_3, B_3^2)$$

with no nonconstant fixed factor? Here degree three is resonant, so the invariant-gap argument does not apply.

APPENDIX B. BOUNDED-DEGREE SCHEMES AND THE SHARP EQUIVARIANT THRESHOLD

This appendix fixes the finite-algebra framework. It also records the exact classification for weights $(-1, 1, 2)$. These calculations originally motivated the unrestricted quartic analysis in the body of the paper.

B.1. The coefficient scheme and collision slice. Fix n, D . After translating the source and target and normalizing the linear part, write

$$F = X + \sum_{d=2}^D H_d.$$

Equating the nonconstant coefficients of $\det JF - 1$ defines a finite-type coefficient scheme $K_{n,D}$. At a Keller map F , its tangent equation in the direction H is

$$\mathrm{tr}((JF)^{-1}JH) = 0$$

coefficientwise.

If a Keller map has a collision, affine source and target transformations place the two source points at $0, e_1$, their common image at 0 , and the first Jacobian at the identity, without increasing ordinary degree. Thus a fixed-degree counterexample search can be conducted in the finite closed incidence slice

$$F(0) = F(e_1) = 0, \quad JF(0) = I, \quad \det JF = 1.$$

This is an implementation framework rather than a novelty claim about finite coefficient loci themselves.

B.2. Equivariant quotient equation. Let

$$\lambda \cdot (x, y, z) = (\lambda^{-1}x, \lambda y, \lambda^2 z), \quad u = xy, \quad v = x^2 z.$$

Every normalized equivariant map has a unique Laurent presentation

$$X = xf(u, v), \quad Y = x^{-1}g(u, v), \quad Z = x^{-2}h(u, v).$$

Polynomiality imposes finite support conditions on f, g, h . The Keller equation is the planar identity

$$gJ(f, h) - 2hJ(f, g) + fJ(g, h) = 1, \quad (\text{B.1})$$

where J denotes the Jacobian in u, v .

Lemma B.1 (Boundary equation). *Let $C_f = V(f)$. Then C_f is smooth and*

$$gD_f h - 2hD_f g = 1 \quad \text{on } C_f, \quad D_f = J(f, -).$$

On an affine-line component, if this is written

$$g_0 h'_0 - 2h_0 g'_0 = \kappa \neq 0,$$

then $\deg g_0 \leq 1$.

Proof. At a singular point of $f = 0$, the left side of (B.1) would vanish. For the degree assertion, compare the leading term. If cancellation occurs because $\deg h_0 = 2 \deg g_0$, subtract a multiple of g_0^2 from h_0 ; this preserves the equation and lowers the degree. Iteration leaves $\deg g_0 \leq 1$. $\square$

Theorem B.2 (Exact equivariant threshold). *Every normalized $(-1, 1, 2)$ -equivariant Keller map of ordinary degree at most six is a polynomial automorphism. The known degree-seven counterexample has this symmetry. Hence the exact minimum in this equivariant class is seven.*

Proof and certificate structure. For degree at most five, the exhaustive support is

$$\begin{aligned} f &= 1 + au + bv + cu^2, \\ g &= u + pv + qu^2 + ruv + su^3 + tv^2, \\ h &= v + Au^2 + Buv + Cu^3 + Dv^2 + Eu^2v. \end{aligned}$$

Lemma B.1 reduces the nonconstant- f locus to four exact coefficient ideals. After the necessary saturations, their Gröbner bases over $\mathbb{Q}$ are all (1). Thus $f = 1$, and (B.1) reduces to $J(g, h) = 1$ for a plane map of degree at most three. It follows that the three-dimensional map is birational and hence an automorphism.

For degree at most six, the exhaustive support is

$$\begin{aligned} f &= 1 + au + bv + cu^2 + duv, \\ g &= u + pv + qu^2 + ruv + su^3 + tv^2 + ku^2v, \\ h &= v + Au^2 + Buv + Cu^3 + Dv^2 + Eu^2v + Fu^4 + Guv^2. \end{aligned}$$

When $d \neq 0$, normalize $d = 1$ and $b \in \{0, 1\}$. Smoothness is the nonvanishing of

$$\Delta = 1 - ab + cb^2,$$

and both ideals saturated by Δ are unit ideals over $\mathbb{Q}$.

When $d = 0, b \neq 0$, normalize $b = 1$. The boundary lemma gives the necessary relations among q, r, s, t, k ; the five resulting charts, including the previously easy-to-miss $c = 0, k \neq 0$ chart, all have unit Gröbner ideals.

Finally suppose $b = d = 0$, so $f = f(u)$. If f is nonconstant, write

$$g = n(u) + m(u)v, \quad h = L(u) + j(u)v + N(u)v^2.$$

With

$$U = fg, \quad V = f^2h,$$

equation (B.1) becomes

$$J(U, V) = f^2.$$

Eliminating v shows that the U^2 - and U -coefficients of V must be constant and that

$$\left. \frac{dV}{du} \right|_U = -\frac{f}{m}.$$

The cases m nonconstant, m a nonzero constant, and $m = 0$ each force the impossible divisibility $f \mid u$. Therefore $f = 1$, and the remaining plane map has degree at most four and is an automorphism.

The finite branches and unit-ideal certificates were generated over $\mathbb{Q}$ and rerun from the recovered exact scripts. Finite-field scans were useful diagnostics but are not logical inputs to the theorem. $\square$

Remark B.3 (A corrected boundary warning). The Laurent boundary equation alone does not force a monomial solution. There are explicit nonmonomial Laurent pairs satisfying it. The proof above uses the complete lifted Keller identity, not that false intermediate shortcut.

B.3. The unrestricted quartic coefficient problem. For normalized maps $\mathbb{A}^3 \rightarrow \mathbb{A}^3$ of ordinary degree at most four, the unrestricted collision coefficient system has 93 variables and 219 nonconstant determinant-coefficient equations. This count is exact, but it does not by itself make the system tractable.

Inside the weight- $(1, -1, -2)$ quartic subfamily, exact coefficient elimination makes every Keller map tame. The calculation is a useful negative result for that symmetry class; it is not a classification of unrestricted quartic Keller maps. In particular, no lower bound for a general counterexample follows merely from this computation.

Question B.4 (Degrees five and six). *Can the unrestricted degree-five and degree-six collision schemes be reduced to finitely many invariant leading-form strata with certificates comparable to those above? More specifically, which parts of the leading-curve, maximal-contact, and Keller-jet analysis used in degree four persist before the degree-seven example appears?*

ACKNOWLEDGMENTS AND DISCLOSURE

The discovery attribution for the counterexample is recorded in the introduction. The author credits Angelo Vistoli, Harry Appelgate, Hironori Onishi, Masayoshi Nagata, Michiel de Bondt, Arno van den Essen, and C. T. C. Wall for the prior results used to delimit and prove the statements above.

AI systems were used extensively for exploration, proof drafting, symbolic code, literature triage, and manuscript editing. They are not authors. The human author is responsible for every mathematical claim. Exact symbolic checks are identified in section 9; they do not replace review of the geometric reductions.

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